MERIT EDUCATION GROUP

Scientific Notation

I can write numbers in scientific notation and perform arithmetic in scientific notation.

Standard: 9.3.5.1

Day 5 · Unit 1 · Theme: Itasca State Park

1 BELL RINGER — WHITEBOARDS READY!

Write 4,000,000 using a power of ten, then write 3.2×10^4 in standard form.

Your response: ______________________

2 KEY VOCABULARY

Scientific Notation — A way to write very large or very small numbers as a coefficient times a _________ of ten. __________________

Coefficient (Sci. Not.) — The number in front of the power of ten, written between 1 and _________.

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Power of Ten — 10 raised to an exponent; 10^6 has a _________ of 6 zeros when written out.

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3 DIRECT INSTRUCTION — TAKE NOTES (I DO)

Example: Write 24,000 in scientific notation, then evaluate 6.1×10^3 in standard form.

• Step 1 — Move the decimal to leave one nonzero digit: 2.4, and count the places moved _________. (4)

• Step 2 — Write as a coefficient times a power of ten: 2.4 × _________. (10^4)

• Step 3 — For 6.1 × 10^3, move the decimal 3 places right: _________. (6,100)

4 GUIDED PRACTICE — WORK TOGETHER (WE DO)

• Write 0.00052 using scientific notation. Multiply 2 x 10^3 by 3 x 10^4.

5 CULTURE CONNECTION — MINNESOTA STORIES

Ethiopian (Ethiopia) — Ethiopian Americans are a growing Minnesota community, centered in Cedar-Riverside. Ethiopian academic tradition dates to antiquity, including the Ge'ez numeral system used for counting and record-keeping.

Word problem: An Ethiopian scribe records the number of star-grain seeds counted in Ge'ez as 4 x 10^8. Write 4 x 10^8 in standard form, and state how many times larger it is than 4 x 10^3.

6 INDEPENDENT PRACTICE — YOU TRY!

• A cell phone has 6.4×10^{10} bytes of storage. Write it in standard form.

7 EXIT TICKET — SHOW WHAT YOU KNOW

Write 0.00052 using scientific notation, then write 2×10^{7} in standard form.

Answer: ____________________

Show your work for full credit.